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ENBC 332

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12/03/2025

Homework 10 Discussion and Outputs

Problem 1:

Source	SS	df	MS	F
Between Groups	750	3	250	5
Within Groups	1800	36	50	
Total	2550	39		

Problem 2:

The null hypothesis is that there is no difference between all of the treatment group means. The alternative hypothesis asserts that at least one group has a different meaning from the others.

Part B

N = 1000

k = 5

$$X_{Grand} \approx -4.1032$$

$$SSB = \sum n_i(X_i - X_{Grand})^2$$

Control	Group A	Group B	Group C	Group D
3041.38	43.71	3741.30	923.21	315.35

SSB = 8064.99

SSW = 23792.67

$$SST = 31857.66$$

$$df_{between} = k - 1 = 5 - 1 = 4$$

$$df_{within} = N - k = 1000 - 5 = 995$$

$$MSB = \frac{SSB}{df_{between}} = \frac{8064.99}{4} = 2016.25$$

$$MSW = \frac{SSW}{df_{within}} = \frac{23792.67}{995} = 23.91$$

$$F = \frac{MSB}{MSW} = \frac{2016.25}{23.91} = 84.32$$

$$F_{crit} = 2.38$$

Source	SS	df	MS	F	F – Crit a = 0.05
Between Groups	8064.99	4	2016.25	84.32	2.38
Within Groups	23792.67	995	23.91		
Total	31857.66	999			

$$F_{stat} > F_{crit} \rightarrow \text{Reject null hypothesis}$$

Part C

$$P_{value} \approx 1.59 \cdot 10^{-134}$$

$$\alpha = 0.05$$

$$P_{value} < \alpha \rightarrow \text{Reject null hypothesis}$$

Part D

Group 1	Group 2	Mean Diff	P-Value	Reject Null?	Conclusion
Control	Treatment A	-4.37	0	Yes	Different A
Control	Treatment B	-8.22	0	Yes	Different B
Control	Treatment C	-1.75	0.003	Yes	Different C
Control	Treatment D	-5.16	0	Yes	Different D
Treatment A	Treatment B	-3.86	0	Yes	Different B
Treatment A	Treatment C	2.62	0	Yes	Different C
Treatment A	Treatment D	-0.79	0.490	No	No sig diff

Treatment B	Treatment C	6.47	0	Yes	Different C
Treatment B	Treatment D	3.07	0	Yes	Different D
Treatment C	Treatment D	-3.4	0	Yes	Different D

Post hoc tests are used to determine which group has a different meaning. This is especially useful due to the ANOVA test results not being able to provide information about this.

The following conclusions can be made based on this test:

- Treatment B is the most effective with the largest negative mean difference compared to the control.
- All treatments beat the control
- Treatment A and treatment D are statistically similar

Part F

1. It is because ANOVA is able to compare multiple groups at the same time. For example, if you have five groups and want to find out which group is different, you will need to run ten separate t-tests. As a result, the computational heaviness increases linearly with the number of groups. On the other hand, if you just use the ANOVA test, you will only need to run it once.
2. The F-statistic is the signal to noise ratio where the variance between the groups is the signal and is divided by the variance within groups which is the noise.
3. ANOVA is a global test and its purpose is to compare if there are equal means that are not for each group. As a result, it is impossible for us to identify which group actually has a different mean. For this reason, the post hoc test must be performed to identify which group has a different mean.
4. Standard ANOVA assumes that all groups have the same amount of variance because if violated, the pooled variance calculated using each group's variance becomes inaccurate and resulting in unreliable conclusions from the f test.
5. Statistical significance confirms by outputting a low p value that the difference in blood pressure reduction is real and not due to random chance. Practical or clinical significance asks whether that difference is large enough to improve a patient's health or change a doctor's treatment plan. Within this dataset that we analyzed, treatment B is likely both where the difference is statistically undeniable, and the 8-mmHg reduction is substantial enough to meaningfully benefit a patient's heart health.

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> # Christopher A. Lee
> # ENBC 332
> # Homework 10 Problem 2
> # 12/03/2025
>
> # Given data
> install.packages("readxl")
trying URL 'https://cran.rstudio.com/bin/macosx/big-sur-arm64/contrib/4.5/readxl_1.4.5.tgz'
Content type 'application/x-gzip' length 1091570 bytes (1.0 MB)
=====
downloaded 1.0 MB

The downloaded binary packages are in
  /var/folders/_t/nzsvtw451k30mrn3mn6c1vbc0000gp/T//RtmpdKULcf/downloaded_packages
> library(readxl)
> df <- read_excel('/Users/christopherlee/Desktop/R Workspace/ENBC 332 - Homework10/medical_dataset.xlsx', sheet = "Sheet1")
>
> anova_model <- aov(Reduction_mmHg ~ Group, data = df)
> summary(anova_model)
              Df Sum Sq Mean Sq F value Pr(>F)
Group           4   8065   2016.2    84.32 <2e-16 ***
Residuals     995  23793     23.9
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>

```